Calculating with Vectors

Exam code: 0625 & 0972

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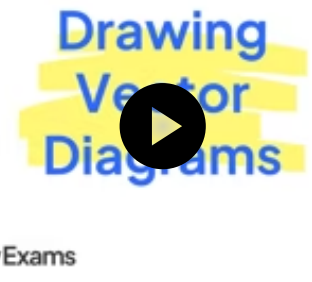

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Help me understand

Calculations with vectors

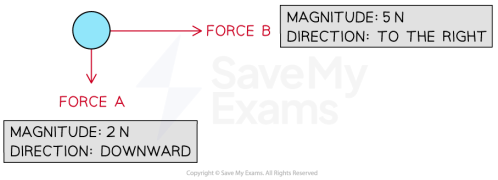
Extended tier only

- **Vectors** can be drawn using **vector diagrams**

Vector diagrams

- **Vectors** are represented by an arrow
  - The **length** of the arrow represents the **magnitude**
  - The **direction** of the arrow indicates the **direction**
  - the **scale** of the arrows should be **proportional** to the relative magnitudes of the forces
    - an arrow for a 4 N force should be twice as long as an arrow for a 2 N force

Vector diagram of two forces acting on an object



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

The length of the arrows are proportional to the magnitude of the forces, and show the direction that forces act in

- Vectors at **right angles** to one another can be **combined** into one **resultant vector**
  - The resultant vector will have the combined effect of the two original vectors
  - For example, a resultant force vector will have the combined effect of two component forces

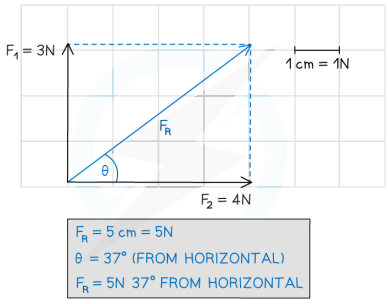
- **Component** vectors are sometimes drawn with a dotted line and a **subscript** indicating horizontal or vertical
  - A force F, for example, may have two components:
    - F_V is the vertical component of the force F
    - F_H is the horizontal component of force F

- To calculate vectors graphically means carefully producing a scale drawing with all lengths and angles correct
  - This should be done using a sharp **pencil**, **ruler** and **protractor**

- Follow these steps to carry out calculations with vectors on graphs

1. Choose a **scale** which fits the page
   - For example, use 1 cm = 10 m or 1 cm = 1 N, so that the diagram is around 10 cm high

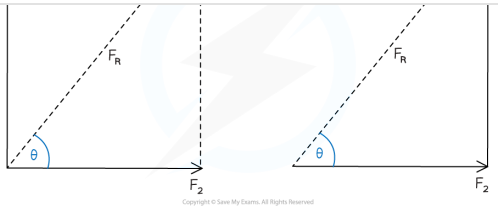
2. Draw the vectors at **right angles** to one another
3. Complete the rectangle
4. Draw the **resultant vector** diagonally from the origin
5. Carefully **measure the length** of the resultant vector
6. Use the scale factor to calculate the **magnitude**
7. Use the protractor to measure the **angle**



Vectors can be measured or calculated graphically using scaled vector diagrams

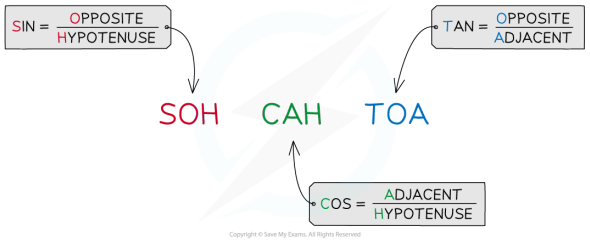
Combining vectors by calculation

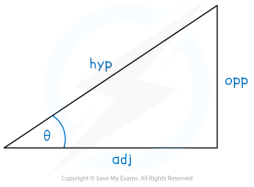
- In this method, a vector diagram is still essential but it does **not** need to be exactly to scale
- The vector diagram can take the form of a sketch, as long as the resultant side, component sides are clearly labelled



Using a vector diagram to resolve two force vectors F_1 and F_2 into a resultant force vector F_R

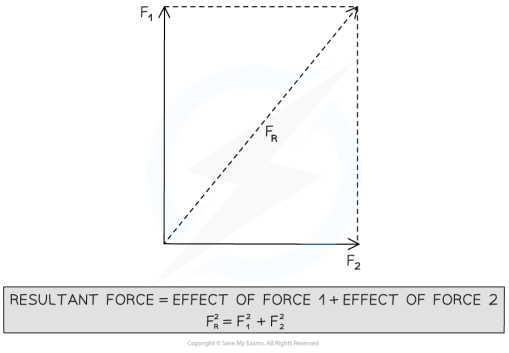
- When the magnitude of only one vector is known, **and** the angle is known, then **trigonometry** can be used to find the magnitude of the missing vector
  - The mnemonic '**soh-cah-toa**' can used to remember the trigonometric functions

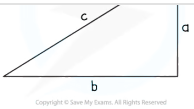




Trigonometry can be used when the magnitude of one vector and the angle is known

- When the magnitudes of two of the vectors are known, then **Pythagoras' theorem** can be used to find the magnitude of the missing vector





Pythagoras's theorem can be used when the magnitudes of two of the three vectors are known

Worked Example

A force acts on an object with 60 N to the right. A second force of 100 N acts on the same object in the upward direction.

Calculate the resultant force acting on the object.

Answer:

Step 1: Draw a vector diagram

Step 2: Calculate the magnitude of the resultant force using Pythagoras' theorem

$$F = \sqrt{60^2 + 100^2}$$

$$F = \sqrt{13\,600}$$

$$F = 117 \text{ N}$$

Step 3: Calculate the direction of the resultant vector using trigonometry

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\tan\theta = \frac{100}{60}$$

$$\theta = \tan^{-1}\left(\frac{100}{60}\right) = 59°$$

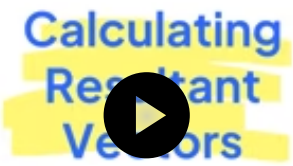
Step 4: State the final answer, complete with magnitude and direction

$F = 117\text{ N at } 59°$ from the horizontal

Examiner Tips and Tricks

If the question specifically asks you to use the calculation or graphical method, you must solve the problem as asked. However, if the choice is left up to you then any correct method will lead to the correct answer.

The graphical method sometimes feels easier than calculating, but once you are confident with trigonometry and Pythagoras you will find calculating quicker and more accurate.



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